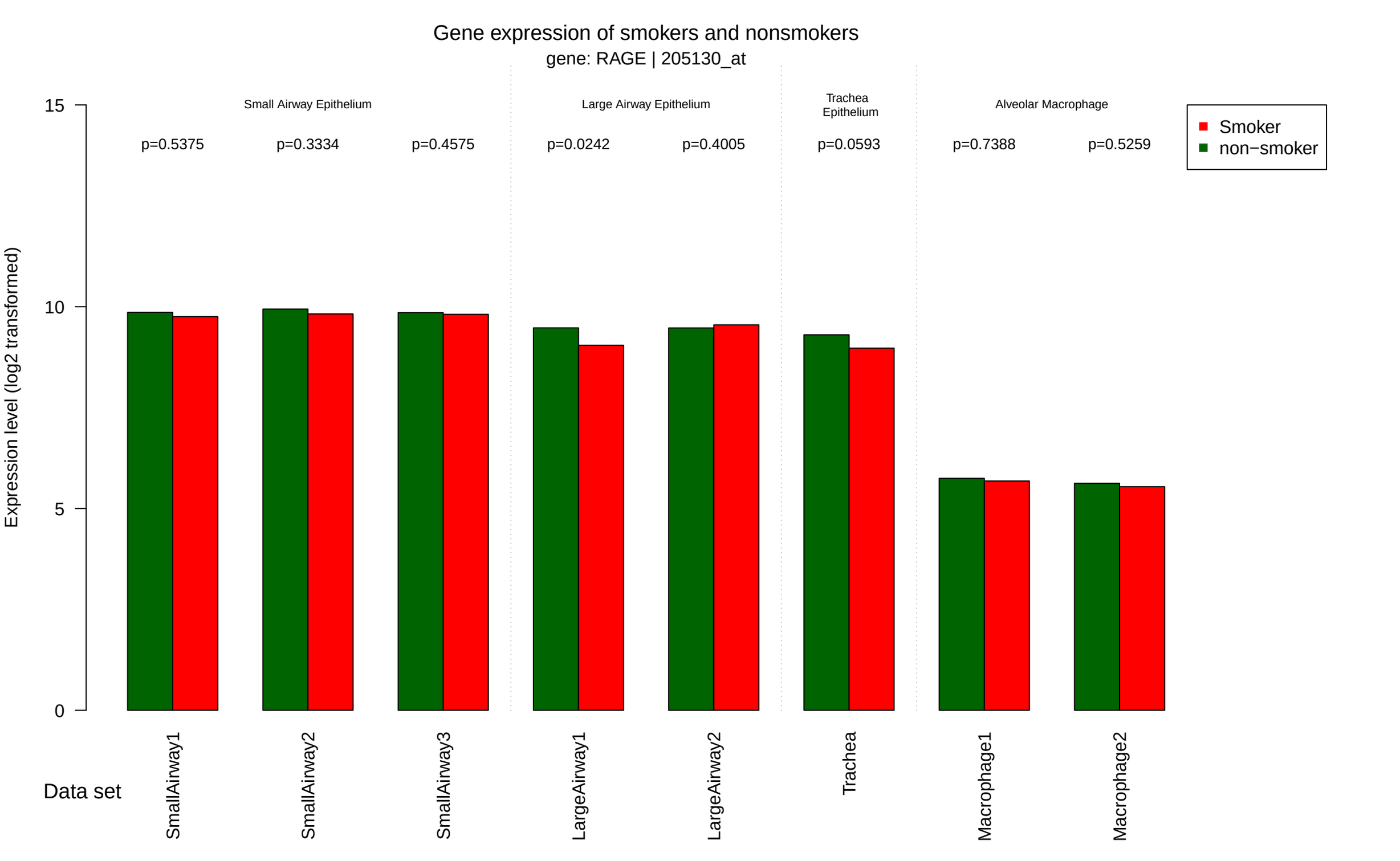
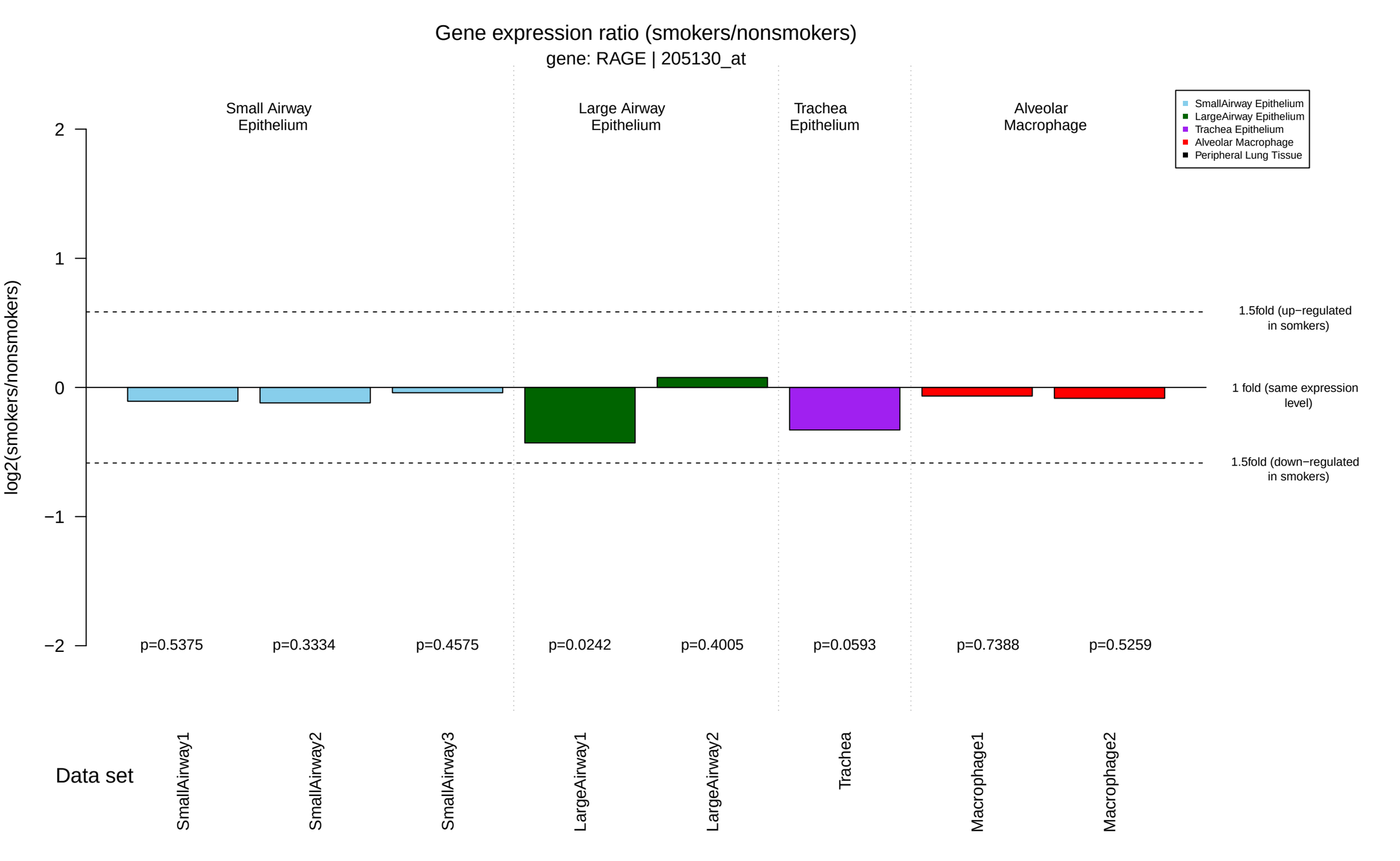
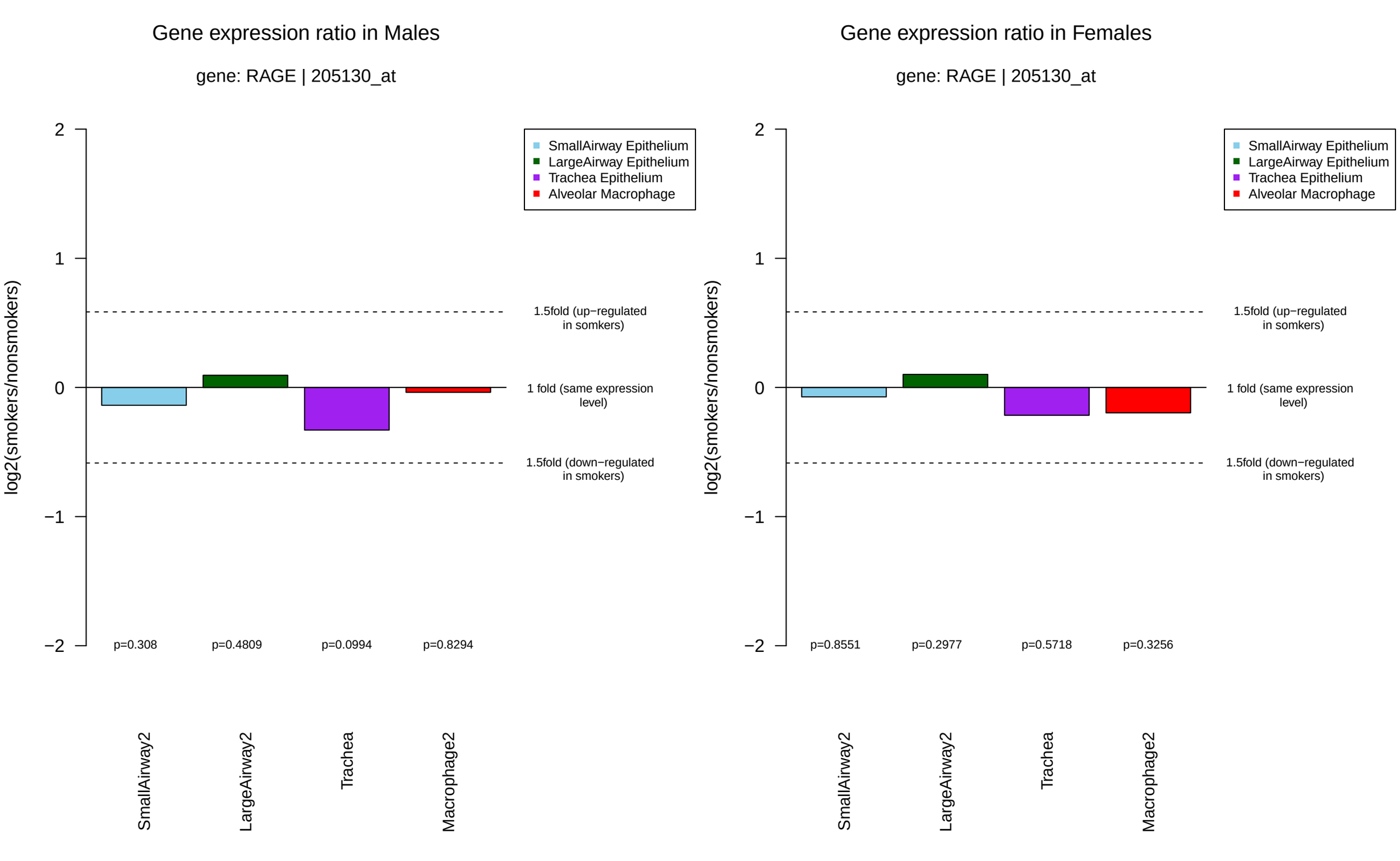
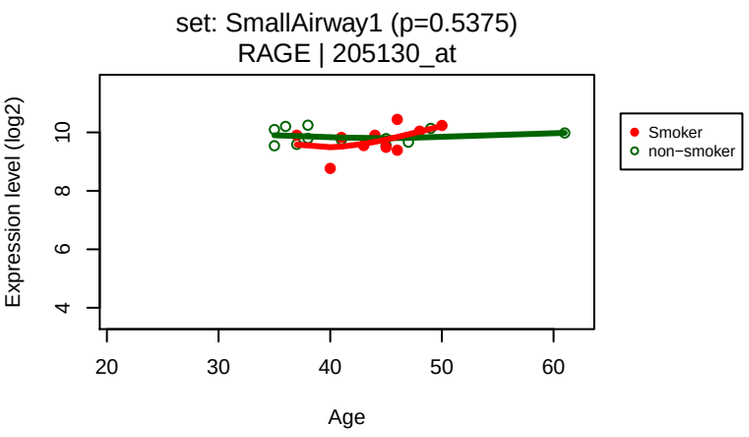
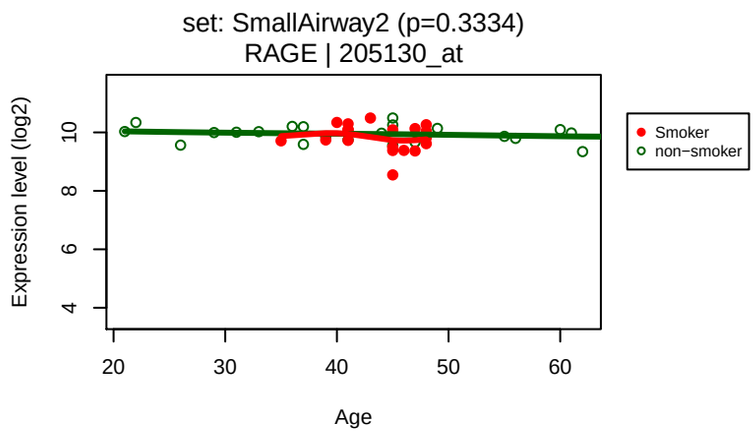
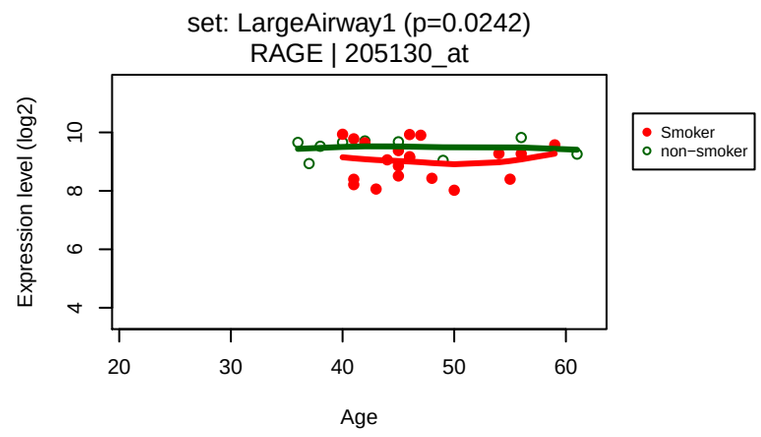

Gene expression of smokers and nonsmokers
gene: RAGE | 205130_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5375
p=0.3334
p=0.4575
p=0.0242
p=0.4005
p=0.0593
p=0.7388
p=0.5259
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

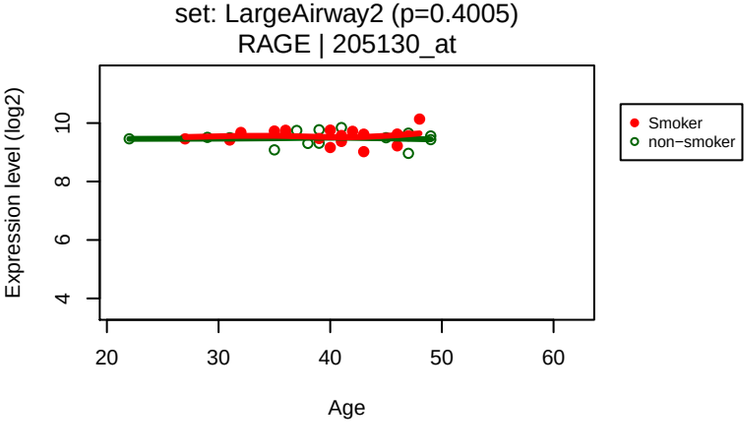

Gene expression ratio (smokers/nonsmokers)
gene: RAGE | 205130_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5375
p=0.3334
p=0.4575
p=0.0242
p=0.4005
p=0.0593
p=0.7388
p=0.5259
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

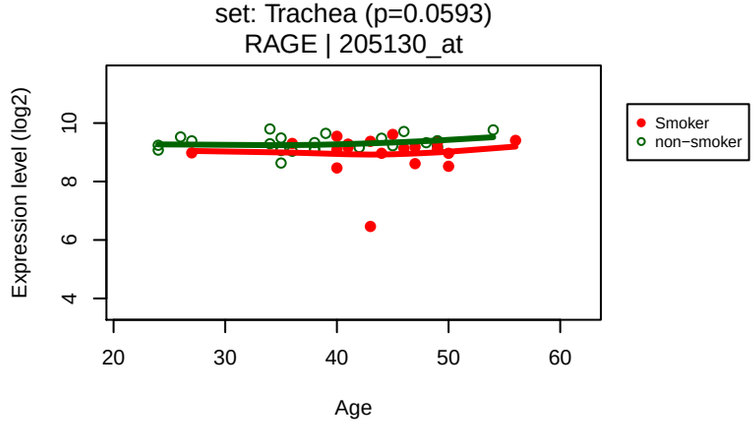

Gene expression ratio in Males
gene: RAGE | 205130_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.308
p=0.4809
p=0.0994
p=0.8294
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAGE | 205130_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8551
p=0.2977
p=0.5718
p=0.3256
SmallAirway2
LargeAirway2
Trachea
Macrophage2

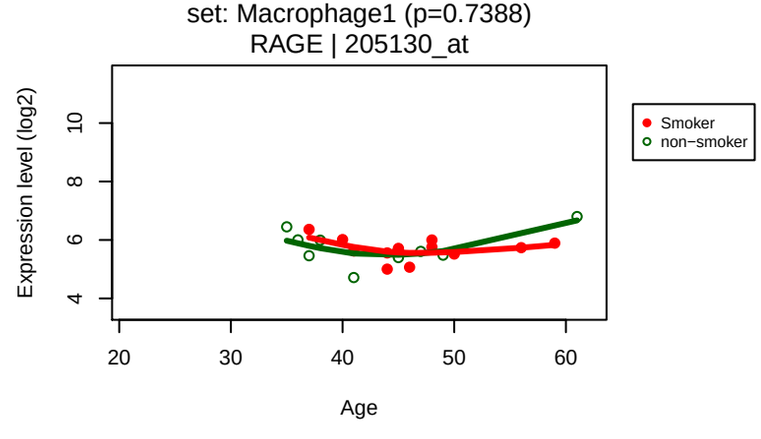

set: SmallAirway1 (p=0.5375)
RAGE | 205130_at
Expression level (log2)
Age
Smoker
non−smoker

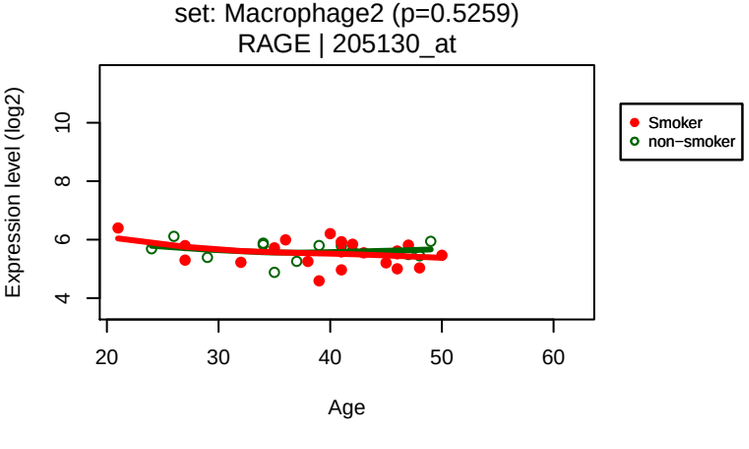

set: SmallAirway2 (p=0.3334)
RAGE | 205130_at
Expression level (log2)
Age
Smoker
non−smoker

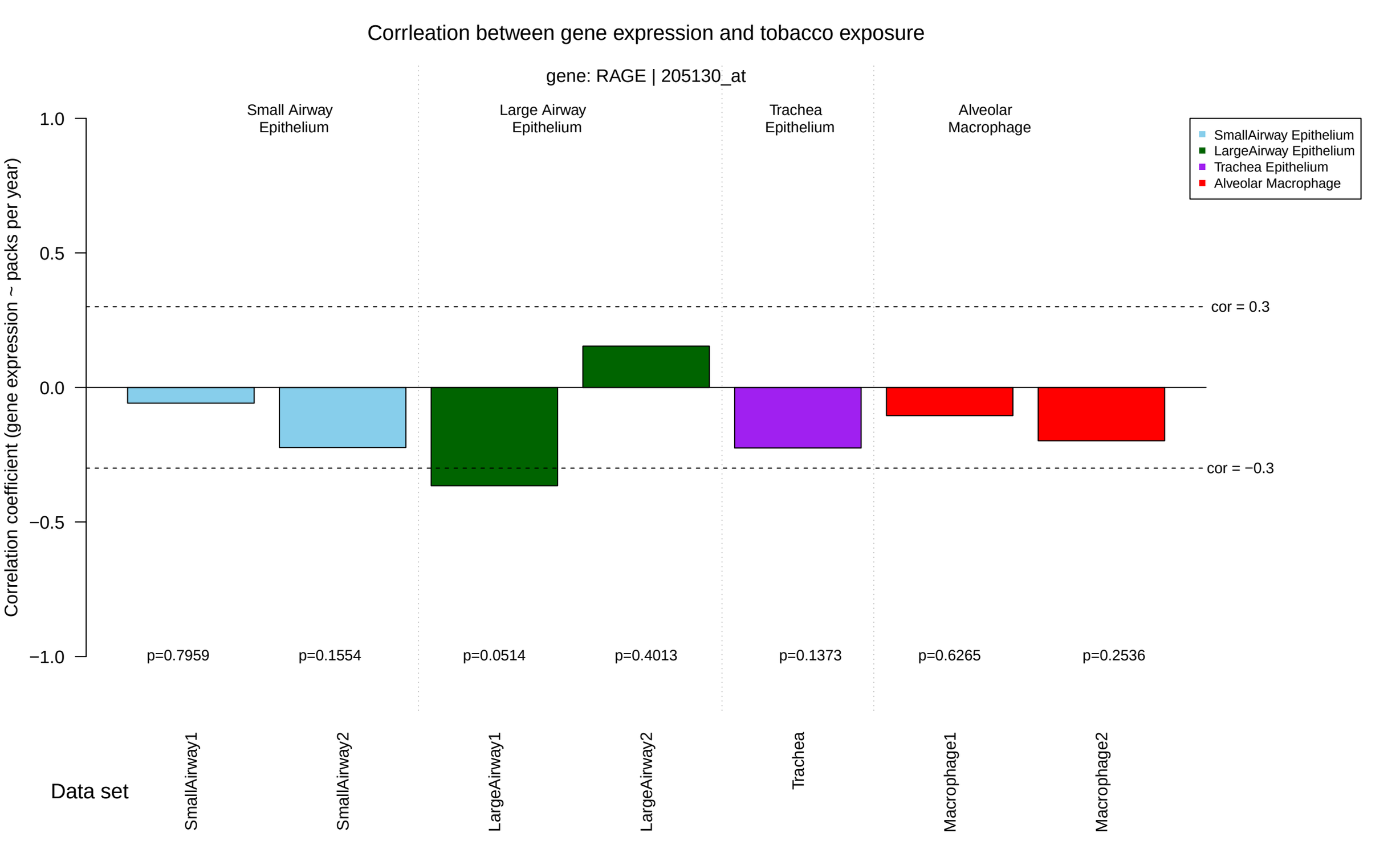

set: LargeAirway1 (p=0.0242)
RAGE | 205130_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4005)
RAGE | 205130_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0593)
RAGE | 205130_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7388)
RAGE | 205130_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5259)
RAGE | 205130_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAGE | 205130_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7959
p=0.1554
p=0.0514
p=0.4013
p=0.1373
p=0.6265
p=0.2536
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2